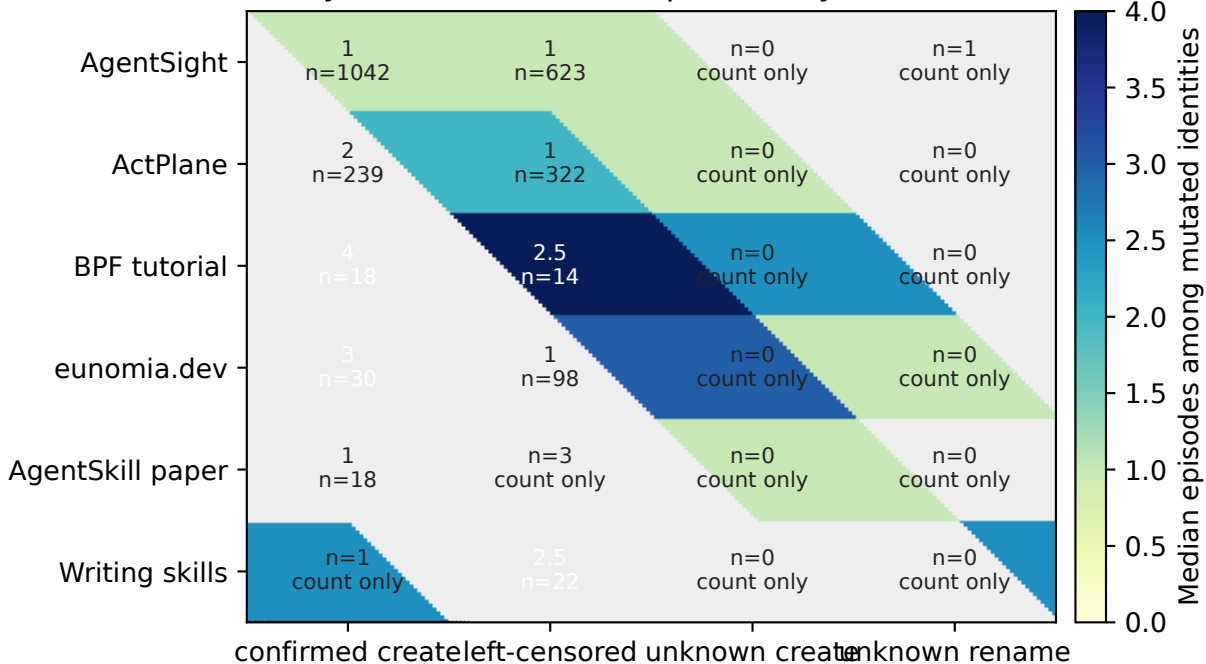


RQ3 sensitivity: median mutation episodes by observable birth state



Cells with fewer than 10 mutated identities remain count-only; unknown birth categories are retained rather than dropped.